Ω AQUASENSE

AquaSense

EO Observation & Few-Shot Classifier

Real-time wetland hydrology intelligence from Sentinel-2 NDWI temporal quantification • Gemini ecological reasoning • Planetary Computer STAC

MISSION

Monitoring Earth's most threatened ecosystems with satellite intelligence

STAC • EO-HYDRO • NDWI
(B03-B08)/(B03+B08)
10m/px Resolution

PALLIKARANAI MARSH • CHENNAI
Baseline 2019 → Target 2025
Cloud <20% • Threshold >0.20

The Wetland Crisis

WORLDWIDE

35% LOST

Of natural wetlands lost since 1970.
Fastest disappearing ecosystem.
Ramsar: 3x faster than forests.

MONITORING GAP

Data Blindness

No real-time, high-res monitoring
for local conservation bodies.
Manual surveys are slow.

NO EARLY WARNI

Climate Risk

Urban encroachment, precipitation
anomalies go undetected until
irreversible. Chennai -34%.

We need an autonomous, satellite-native observatory that turns raw spectral bands into actionable ecological intelligence.

Introducing AquaSense

An end-to-end EO pipeline that ingests Sentinel-2, computes NDWI, and delivers hydrological change AI synthesis.

- STAC Auto-Discovery — Queries Planetary Computer for best low-cloud scenes with retry-on-failure
- NDWI Intelligence — (B03-B08)/(B03+B08) with dynamic thresholding & 5 color ramps
- Pixel Quantification — Real-time water extent in km², % change, longitudinal trend
- Gemini Ecological AI — 4 modes: Deep Reasoning, Search, Maps, Fast Summary
- Interactive Observatory — Drag BBOX anchors, split-slider, diff mask
- Field Verification — Multimodal image understanding for ground-truthing

01 **STAC INGEST**

Sentinel-2 L2A
Planetary Computer
Cloud <20%

02 **NDWI COMPUTE**

(B03-B08)/(B03+B08)
Colorized LUT
Threshold >0.20

03 **DIFF & TREND**

Water pixels x0.0001
km² change
Annual snapshots

04 **AI SYNTHESIS**

Gemini 3.7 Flash
Ecology + Search
+ Maps Grounding

System Architecture & Tech Stack

FRONTEND	BACKEND	GEOSPATIAL	ML / AI	GEN AI
React 19 + Vite Tailwind 4.1 Map Editor, Split Slider Recharts, Turf.js	Express + TSX STAC API Client Raster Pipeline Provenance Export	Sentinel-2 L2A NDWI: B03 & B08 Planetary Computer Cloud Filter + BBOX	Clay Encoder Prithvi Foundation Few-Shot Classifier KNN / Logistic	Gemini 3.7 Flash Deep Reasoning Search Grounding Maps + Multimodal

DATA FLOW: User BBOX → STAC Search (5 candidates) → Verify Rasters → NDWI Thresholding → Color LUT → Diff Mask → Trend → Gemini → Export

✔ Sentinel-2 10m resolution (0.0001 km²/pixel) | ✔ Microsoft Planetary Computer STAC v1 | ✔ NDWI = (Green - NIR)/(Green + NIR)

✔ Retry-On-Failure: Tests top 5 scenes for raster integrity | ✔ Real-time threshold recalculation

05 / FEATURES

Observatory-Grade Features

Interactive BBOX Editor

Drag corner anchors on map, upload GeoJSON, live telemetry. Prec

Dynamic NDWI Controls

Slider -0.30 to +0.70, presets. Instant recalc of km² water extent & tr

5 Scientific Color Ramps

Viridis, Plasma, Inferno, Cividis, Turbo. LUT needle bar, scale legend.

True Color vs NDWI Swipe

Split-slider with handle, T0 vs T1, Raw vs Colorized, Diff Mask.

Cloud Filter Intelligence

Slider 1-80%, presets. STAC query sorts by eo:cloud_cover ASC.

Provenance & Export

JSON with timestamp, hash, methodology, bbox, quantification, scene IDs.

Few-Shot Intelligence & Gemini Reasoning

Beyond NDWI: We embed spectral patches and classify with minimal labels — then explain it with grounded LLMs.

ENCODERS: Clay + Prithvi
CLASSIFIERS: KNN, Logistic, Few-Shot
PATCH EXTRACTION: Sliding window
FINE-TUNING: PyTorch adapters

GEMINI MODES:
- Deep Reasoning (3.7 Flash)
- Search Grounding
- Maps Grounding
- Fast Summary (3.1 Lite)
- Multimodal Field Inspection

Four-Mode Ecological Intelligence

DEEP THINKING
gemini-3.7-flash
Hydrological trajectory, driving factors, ecosystem services, conservation.

SEARCH GROUNDED
gemini-3.5-flash + GoogleSearch
Live factual grounding: recent reports, govt actions, climate events.

MAPS GROUNDED
gemini-3.5-flash + GoogleMaps
Geographic orientation: landmarks, protected zones, urban centers.

FAST SUMMARY
gemini-3.1-flash-lite
3-4 bullet low-latency summary for dashboards. Optimized for speed.

FIELD INSPECTION
Multimodal Vision
Upload drone/field photo → water clarity, algae, vegetation + few-shot scores.

Observatory Interface Walkthrough

PIPELINE TELEMETRY

1. Planetary Computer STAC ✓
2. NDWI Raster & LUT ✓
3. Hydro Differencing ✓

[Interactive BBOX Map]
Drag anchors
12.91,80.20 → 12.95,80.23

TRUE COLOR SWIPE | NDWI SWIPE (VIRIDIS) | DIFF MASK

[Satellite Split Slider]
2019 ← drag → 2025
True Color + NDWI Colorized

NDWI Scale: -1.0 [Water ← → Land] +1.0 Threshold >0.20

HYDROLOGICAL QUANTIFICATION

2019 WATER: 4.82 km²
2025 WATER: 3.15 km²
CHANGE: -1.67 km²
RELATIVE: -34.6%

■ GEMINI INSIGHTS

Hydrological Trajectory...
Driving Factors...
Ecosystem Services...
Recommendations...

Real-World Impact & Use Cases

Government & Policy

Wetland Authority monitoring, Ramsar reporting, encroachment dete

Conservation NGOs

Biodiversity assessment, habitat loss alerts, restoration prioritization

Urban Planning

Chennai Pallikaranai: -34% loss correlates with urbanization. Early warning for b

Research & Academia

Longitudinal studies, climate correlation, open STAC data, provenar

Disaster Response

Pre-monsoon capacity, post-flood inundation mapping, drought early

Citizen Science

Field photo upload for ground-truthing, community watch, EO literacy tool.

Scientific Rigor & Quantification

FORMULA: NDWI = (Green - NIR)/(Green + NIR) = (B03-B08)/(B03+B08)

- B03 = 560nm Green, B08 = 842nm NIR, Sentinel-2 L2A
- Range -1 to +1, Water >0.20 typically

PIXEL COUNTING:

- Canvas-based raster analysis (offscreen)
- countWaterPixelsWithThreshold() → water pixels
- Area = pixels × 0.0001 km² (10m × 10m)

DIFFERENCE MASK:

- generateDifferenceMapWithThreshold()
- Blue = Gained, Red = Lost, Dark Blue = Unchanged

ROBUSTNESS:

- Retry-On-Failure across top 5 STAC candidates
- Pre-cache verification with getCachedImage()
- Cloud occlusion advisory >15%

```
pipeline.ts // core logic

POST /api/stac/v1/search
{
  collections: ["sentinel-2-l2a"],
  bbox: [80.20,12.91,80.23,12.95],
  datetime: "2019-03-01/2019-03-31",
  query: { "eo:cloud_cover": {lt: 20} },
  sortby: [{field: "eo:cloud_cover"}],
  limit: 5
}

NDWI URL:
.../preview.png?expression=(B03-B08)/(B03+B08)
&rescale=-1,1

Area:
area_km2 = water_pixels * 0.0001
change = areaB - areaA
pct = (change/areaA)*100

Export:
{
  methodology: "NDWI >0.20, 10m",
  system_hash: "0x8a92f02c",
  quantification: { yearA, yearB }
}
```

Roadmap & Future Vision

NOW - Q3 2025

✓ MVP Observatory
✓ STAC + NDWI Pipeline
✓ Gemini 4-Mode Insights
✓ Pallikaranai Validation

NEXT - Q4 2025

- Multi-index: NDVI, MNDWI
- Time-series animation
- User accounts & saved AOIs
- Mobile PWA + offline

Q1 2026

- Clay & Prithvi fine-tuning U
- Few-shot labeling tool
- Crowdsourced ground-truth
- Model comparison

Q2 2026 - SCALE

- Pan-India wetland atlas
- API for NGOs & govt
- Alert system (loss >10%)
- Partnerships: ISRO, WWF

VISION 2027

- Global wetland watch
- Carbon credit quantification
- Flood prediction
- Open science platform

AquaSense

Protecting wetlands with satellite intelligence.

Every pixel tells a story.
Every km² matters.

Built with:
• Microsoft Planetary Computer
• Sentinel-2 L2A • NDWI Science
• Gemini Ecological Reasoning
• Few-Shot Foundation Models

GET STARTED

Live Demo: aquasense.eo
GitHub: Naeha-S/Aquasense

Try Pallikaranai Marsh
Chennai • 2019 → 2025
BBOX: 80.20,12.91,80.23,12.95

Contact:
[Your Name]
[Email] • [LinkedIn]

"From spectral bands to ecological insights — autonomous, explainable, actionable."